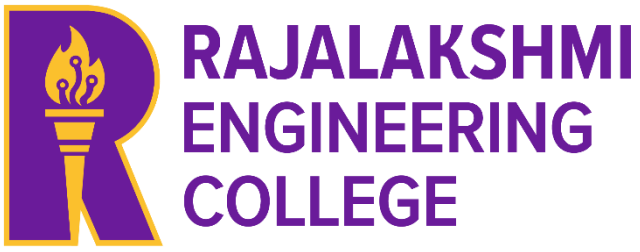
RAJALAKSHMI ENGINEERING COLLEGE

RAJALAKSHMI NAGAR, THANDALAM – 602 105



CS23331 - DESIGN AND ANALYSIS OF ALGORITHMS

Laboratory Record Note Book

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Semester:	:	III
Academic Year :	:	2026 - 2027

GREEDY ALGORITHMS

EXPERIMENT 06

1-G-Coin Problem

Greedy Algorithms

Started on	Tuesday, 8 September 2026, 11:20 AM
State	Finished
Completed on	Tuesday, 8 September 2026, 11:24 AM
Time taken	4 mins 10 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Write a program to take value V and we want to make change for V Rs, and we have infinite supply of each of the denominations in Indian currency, i.e., we have infinite supply of { 1, 2, 5, 10, 20, 50, 100, 500, 1000} valued coins/notes, what is the minimum number of coins and/or notes needed to make the change.

Input Format:

Take an integer from stdin.

Output Format:

print the integer which is change of the number.

Example Input :

64

Output:

4

Explanaton:

We need a 50 Rs note and a 10 Rs note and two 2 rupee coins.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int main()
{
    int v;
    scanf("%d", &v);
    // Array of available Indian Currency denominations sorted in descending order
    int den[]= {1000, 500, 100, 50, 20, 10, 5, 2, 1};

    int n=sizeof(den)/sizeof(den[0]);

    int count=0;

    for (int i=0;i<n;i++)
    {
        if (v==0)
        {
            break;
        }
        if (v>=den[i])
        {
            count+=v/den[i];
            v=v%den[i];
        }
    }
    printf("%d\n",count);
    return 0;
}
```

	Input	Expected	Got	
✔	49	5	5	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

EXPERIMENT 07

2-G-Cookies Problem

Greedy Algorithms

Started on	Tuesday, 8 September 2026, 11:25 AM
State	Finished
Completed on	Tuesday, 8 September 2026, 11:34 AM
Time taken	9 mins 28 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Assume you are an awesome parent and want to give your children some cookies. But, you should give each child at most one cookie.

Each child i has a greed factor g[i], which is the minimum size of a cookie that the child will be content with; and each cookie j has a size s[j]. If s[j] >= g[i], we can assign the cookie j to the child i, and the child i will be content. Your goal is to maximize the number of your content children and output the maximum number.

Example 1:

Input:

3

1 2 3

2

1 1

Output:

1

Explanation: You have 3 children and 2 cookies. The greed factors of 3 children are 1, 2, 3.

And even though you have 2 cookies, since their size is both 1, you could only make the child whose greed factor is 1 content.

You need to output 1.

Constraints:

1 <= g.length <= 3 * 10^4

0 <= s.length <= 3 * 10^4

1 <= g[i], s[j] <= 2^31 - 1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int g[30000];
    for (int i = 0; i < n; i++)
    {
        scanf("%d", &g[i]);
    }

    int m;
    scanf("%d", &m);

    int s[30000];
    for (int i = 0; i < m; i++)
    {
        scanf("%d", &s[i]);
    }

    for (int i = 0; i < n - 1; i++)
    {
        for (int j = 0; j < n - i - 1; j++)
        {
            if (g[j] > g[j + 1])
            {
                int t = g[j];
                g[j] = g[j + 1];
                g[j + 1] = t;
            }
        }
    }

    for (int i = 0; i < m - 1; i++)
    {
        for (int j = 0; j < m - i - 1; j++)
        {
            if (s[j] > s[j + 1])
            {
                int t = s[j];
                s[j] = s[j + 1];
                s[j + 1] = t;
            }
        }
    }

    int i = 0;
    int j = 0;

    while (i < n && j < m)
```

	Input	Expected	Got	
✔	2 1 2 3 1 2 3	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

EXPERIMENT 08

3-G-Burger Problem

Greedy Algorithms

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Monday, 28 September 2026, 10:58 PM
State	Finished
Completed on	Monday, 28 September 2026, 10:59 PM
Time taken	39 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

A person needs to eat burgers. Each burger contains a count of calorie. After eating the burger, the person needs to run a distance to burn out his calories. If he has eaten i burgers with c calories each, then he has to run at least $3^i * c$ kilometers to burn out the calories. For example, if he ate 3 burgers with the count of calorie in the order: [1, 3, 2], the kilometers he needs to run are $(3^0 * 1) + (3^1 * 3) + (3^2 * 2) = 1 + 9 + 18 = 28$. But this is not the minimum, so need to try out other orders of consumption and choose the minimum value. Determine the minimum distance he needs to run. Note: He can eat burger in any order and use an efficient sorting algorithm.Apply greedy approach to solve the problem.

Input Format

First Line contains the number of burgers

Second line contains calories of each burger which is n space-separate integers

Output Format

Print: Minimum number of kilometers needed to run to burn out the calories

Sample Input

```
3
5 10 7
```

Sample Output

```
76
```

For example:

Test	Input	Result
Test Case 1	3 1 3 2	18

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <math.h>
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++)
    scanf("%d",&arr[i]);
    for(int i=0;i<n-1;i++){
        for(int j=0;j<n-1-i;j++){
            if(arr[j]<arr[j+1]){
                int t=arr[j];
                arr[j]=arr[j+1];
                arr[j+1]=t;
            }
        }
    }
    int r=0;
    for(int i=0;i<n;i++){
        int a=pow(n,i);
        r=r+a*arr[i];
    }
    printf("%d",r);
}
```

	Test	Input	Expected	Got	
✔	Test Case 1	3 1 3 2	18	18	✔
✔	Test Case 2	4 7 4 9 6	389	389	✔
✔	Test Case 3	3 5 10 7	76	76	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

EXPERIMENT 09

4-G-Array Sum max problem

Greedy Algorithms

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Tuesday, 8 September 2026, 10:06 PM
State	Finished
Completed on	Tuesday, 8 September 2026, 10:10 PM
Time taken	4 mins 10 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct | Mark 1.00 out of 1.00 | Flag question

Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main()
{
    int n,a[100],temp;
    long long x=0;

    scanf("%d",&n);

    for(int i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }

    for(int i=0;i<n;i++)
    {
        for(int j=i+1;j<n;j++)
        {
            if(a[i]>a[j])
            {
                temp=a[i];
                a[i]=a[j];
                a[j]=temp;
            }
        }
    }

    for(int i=0;i<n;i++)
    {
        x+=a[i]*i;
    }
    printf("%lld",x);
    return 0;
}
```

	Input	Expected	Got	
✔	5 2 5 3 4 0	40	40	✔
✔	10 2 2 2 2 4 4 3 3 5 5 5	191	191	✔
✔	2 45 3	45	45	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

EXPERIMENT 10

5-G-Product of Array elements-Minimum

Greedy Algorithms

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Tuesday, 8 September 2026, 9:47 PM
State	Finished
Completed on	Tuesday, 8 September 2026, 9:57 PM
Time taken	10 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Given two arrays array_One[] and array_Two[] of same size N. We need to first rearrange the arrays such that the sum of the product of pairs(1 element from each) is minimum. That is SUM (A[i] * B[i]) for all i is minimum.

For example:

Input	Result
3 1 2 3 4 5 6	28

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
#include<stdlib.h>

int main()
{
  int n,temp;
  int arr1[100],arr2[100];
  long long sum=0;
  scanf("%d",&n);

  for(int i=0;i<n;i++)
  {
      scanf("%d",&arr1[i]);
  }
  for(int i=0;i<n;i++)
  {
      scanf("%d",&arr2[i]);
  }

  for(int i=0;i<n;i++)
  {
      for(int j=i+1;j<n;j++)
      {
          if(arr1[i]>arr1[j])
          {
              temp=arr1[i];
              arr1[i]=arr1[j];
              arr1[j]=temp;
          }
      }
  }

  for(int i=0;i<n;i++)
  {
      for(int j=i+1;j<n;j++)
      {
          if(arr2[i]<arr2[j])
          {
              temp=arr2[i];
              arr2[i]=arr2[j];
              arr2[j]=temp;
          }
      }
  }

  for(int i=0;i<n;i++)
  {
      sum+=(long long)arr1[i]*arr2[i];
  }
  printf("%lld",sum);
  return 0;
}
```

	Input	Expected	Got	
✔	3 1 2 3 4 5 6	28	28	✔
✔	4 7 5 1 2 1 3 4 1	22	22	✔
✔	5 20 10 30 10 40 8 9 4 3 10	590	590	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.